

Freight Rate Prediction

Approach, results, and lessons learned



This is a forecasting problem

48,000

training loads

January–October 2025

12,000

validation loads

November–December 2025

The model predicts future months from past data. A random split does not represent that task.

Distance is the main pricing signal

Freight rates are strongly related to distance and market conditions.

A Ridge regression performed as well as more complex models in the first experiment.

Baseline Ridge

RMSE 633.9

The coordinates are redundant

0.9995

Correlation between haversine distance and the provided distance

The city coordinates are synthetic.

The provided distance was kept as the main geographic feature.

Validation follows time

05



Target encoding was fitted separately inside each fold.

Log(rate) reduced percentage error

06

Approach	RMSE	MAPE
Baseline Ridge	633.9	8.4%
Ridge with log target	633.0	5.8%

The target transformation improved MAPE without materially changing RMSE.

Complex models reached similar RMSE

07

Model	RMSE
CatBoost	631.6
Ridge	633.2
XGBoost	635.2
LightGBM	635.4

The strongest results fall within a narrow range: 631.6–635.4 RMSE.

Final model

08

Equal-weight average

CatBoost + Ridge + XGBoost

RMSE

631.7

MAE

123.9

MAPE

5.6%

R²

0.825

The average narrowly outperformed the best single model on out-of-fold predictions.

Two prediction files required two models

09

Validation predictions

12,000 rows

Used all available columns and the three-model average.

December predictions

31 daily rows

Used the six shared columns and a reduced CatBoost model.

The model matched the columns available for each output.